Large language models acquire latent representations of human cognitive modes and psychiatric conditions because mental states systematically alter **functional vocabulary, syntactic hierarchy, and semantic graph topology**—features that persist in everyday writing even when authors are not discussing mental health (Self-Reported Mental Health Diagnoses Dataset). Because closed-class function words (pronouns, modal verbs, quantifiers, and prepositions) are processed largely outside conscious awareness (Tausczik & Pennebaker, 2010), they form high-frequency statistical signatures in web-scale pretraining corpora that can be selectively activated through structural prompt design (Patient-Ψ Framework).

How Behavioral Traces Form in Pretraining Data

- **Non-clinical persistence across general topics**: Large-scale corpus analyses such as the SMHD (COLING 2018) Study demonstrate that even after stripping all mental-health subreddits and diagnostic keywords, millions of general-interest posts by users across nine conditions (including 1.27M posts in depression, 872K in ADHD, 795K in anxiety, and 575K in bipolar disorder) remain statistically separable from control corpora based purely on functional word distributions and clausal syntax (SMHD Resource).
- **Style over surface valence**: Computational psycholinguistics using LIWC-22 shows that structural and functional markers track cognitive modes more reliably than topical emotion words. In a study of 63 online forums (>6,400 writers), Al-Mosaiwi & Johnstone (2018) proved that a 19-word dictionary of valence-free **absolutist words** (*completely, nothing, always, entire*) tracked affective severity across anxiety ($d = 3.24$), depression ($d = 3.14$), and suicidal ideation ($d = 4.56$) far more faithfully than negative-emotion lexicons, and remained elevated in depression recovery forums as a persistent cognitive trait (PubMed 30886766).
- **Latent agent state inference**: During autoregressive pretraining, next-token prediction over diverse documents forces the model to infer a latent representation of the generating agent's beliefs, attentional focus, and syntactic constraints (Andreas, 2022). Instruction fine-tuning and parameter scaling consolidate these co-occurring linguistic features into psychometrically valid, internally consistent persona clusters ($\alpha > 0.90$) that can be steered via prompt context (Serapio-García et al., 2025).

Psycholinguistic Signatures and Prompt-Invocation Levers by Mental Mode

Mental Mode / Condition	**Lexical & Functional Word Markers**	**Syntactic & Discourse Topology**	**Prompt Levers to Invoke the Latent Persona**
Depressive / Ruminative Mode (Spruit et al., 2022)	Elevated first-person singular pronouns (*I, me, my*) vs. first-person plural (*we*) (Stirman &	Past-tense anchoring (focuspast), inverted word order, lexical repetition, and	Mandate 1sg subject anchoring, past-tense temporal framing, short low-energy clauses, and

	Pennebaker, 2001); high absolutist quantifiers (*never, nothing, completely*) (PMC6376956).	short, detached, paratactic sentences (Spruit et al., 2022).	unqualified totality adverbs without explicit melodrama.
Anxious / Hypervigilant Mode (Spruit et al., 2022)	High rate of tentative hedges (*maybe, perhaps, might*), impersonal pronouns, and somatic/health monitoring terms (Spruit et al., 2022).	Future-oriented conditional branching (*what if / unless*), impaired inter-sentence semantic coherence under threat scanning, and high modal verb density.	Specify recursive contingency checking, future-conditional clauses, frequent epistemic hedging paired with catastrophic boundary assumptions.
Obsessive-Compulsive (OCD) Mode (Reddit r/OCD Analysis)	Elevated cognitive-mechanism vocabulary (*cause, know, ought*), discrepancy/inhibition terms, and certainty-seeking qualifiers (Spruit et al., 2022).	Exhaustive parenthetical self-correction, recursive "just-right" definitional caveats, and nested conditional clauses aimed at eliminating ambiguity (r/PhD Language Obsessions).	Instruct the persona to preemptively close every interpretive loophole, qualify assertions with recursive parentheticals, and verify causal chains step-by-step.
ADHD / Hyper-Associative Mode (CLPsych 2026)	Fewer relativity tokens, elevated third-person plural (3pl) pronouns, and rapid shifts in concrete topical nouns (Spruit et al., 2022).	Higher sentence count with fewer subordinate clauses per sentence (PubMed 19928595); mid-thought tangential branching (tangent style in Patient-Ψ)	Direct the model to follow immediate semantic associations mid-sentence via dashes or parentheticals, drop transitional scaffolding, and pivot rapidly across

		before looping back.	subtopics.
Autistic (ASC) / Systemizing Mode (Schroeder et al., 2023)	High lexical specificity or idiosyncratic non-relational content choices; reduced pragmatic/social-grooming fillers (PMC9075340).	Sharp reduction in embedded relative clauses (16.8% vs. 42.9% in controls, $p = .0001$), preference for explicit definite noun phrases over ambiguous anaphoric pronouns (PMC10123896).	Require literal, non-idiomatic precision, explicit noun-phrase repetition instead of pronominal shorthand, and flat, coordinate clausal structures.
Manic / Flight-of-Ideas Mode (r/BipolarFlightOfIdeas)	High lexical density, grandiosity/achievement drives, intensified action verbs, and phonetic/clang or polysemous wordplay (r/bipolar).	Pressured, run-on parataxis where sentences chain via surface-level semantic or phonetic bridges rather than global topic hierarchy (Spruit et al., 2022).	Enforce high token velocity, long multi-clause chaining, uninhibited certainty, and associative leaps triggered by the last concept in each clause.
Schizotypal / Formal Thought Disorder (FTD) (Nettekoven et al., 2023)	Elevated 3pl (*they*) and indefinite pronouns, perceptual verbs (*hear, feel*), causal/insight markers, and reduced articles/prepositions (Birnbaum et al., 2017).	Fragmented semantic speech networks with more disconnected sub-components ($p = .046$) and smaller median component size ($p < .01$) (PMC10031728); decay in Latent Semantic Analysis coherence (Elvevåg et al., 2007).	Constrain adjacent sentences to share oblique or metaphorical predicates rather than direct referential continuity, using unanchored pronouns (*they, it*) and apophenic causal links.

Borderline (BPD) / Dichotomous Mode (Al-Mosaiwi & Johnstone, 2018)	High third-person singular (3sg) pronoun focus, elevated swearing/dysphoric intensifiers, and extreme absolutist polarization (Spruit et al., 2022).	Rapid oscillation between idealization and devaluation without cognitive-affective integration words (PMC6376956).	Anchor attention on a specific interpersonal target (3sg), prohibit moderate qualifiers (*somewhat, fairly*), and alternate between totalizing praise and rejection.

Principles for Designing Prompts That Invoke Authentic Personas

- **Avoid diagnostic labels in favor of cognitive conceptualization**: In clinical simulation benchmarks, prompting GPT-4 with a bare diagnostic label (e.g., *"Act as a depressed and anxious patient"*) fails because RLHF post-training causes the model to output clinical textbook summaries—producing a persona that clinicians described as *"almost like doing therapy with a therapist"* (Wang et al., 2024).
- **Program the latent Cognitive Conceptualization Diagram (CCD)**: The Patient-Ψ Framework (EMNLP 2024) proved that feeding an LLM a structured eight-part cognitive schema—comprising **Relevant History**, **Core Beliefs** (e.g., *helpless, unlovable, worthless*), **Intermediate Rules** (*"If I don't control every detail, everything collapses"*), **Coping Strategies**, **Activating Trigger**, and **Automatic Thoughts**—outperformed diagnostic baselines across maladaptive cognition fidelity ($p < .05$) and emotional state realism ($p < 10^{-4}$) (arXiv:2405.19660v2).
- **Orthogonalize cognitive content and conversational delivery style**: Real-world mental states manifest through distinct pragmatic channels. Coupling a cognitive schema with an explicit **conversational style constraint**—such as reserved (minimal, evasive, high prompting threshold), tangent (associative derailment), upset (defensive, resistant to reframing), verbose (over-elaborative), or pleasing (compulsive agreement and symptom minimization)—yielded the single largest fidelity gain ($p < 10^{-4}$) in expert blind evaluations (Patient-Ψ Conversational Styles).
- **Control functional-word budgets and syntactic geometry directly**: Because LLMs represent personas as conditional probability distributions over token sequences (Shanahan et al., 2023), the cleanest way to invoke a cognitive mode without triggering safety refusals or theatrical caricature is to specify **syntactic and lexical constraints**:
  - **Pronoun & tense ratios**: Explicitly set the pronoun anchor (1sg self-focus vs. 3sg/3pl externalized vigilance) and temporal horizon (past-ruminative vs. future-conditional) (Spruit et al., 2022).
  - **Epistemic modality**: Dial between the 19-word absolutist lexicon (*always, nothing, completely, entirely*) for rigid/polarized states (PMC6376956) and dense

tentative/discrepancy markers (*maybe, ought, unless, technically*) for anxious or obsessive checking (Birnbaum et al., 2017).

- **Clausal embedding & graph connectivity**: Constrain sentence architecture—such as banning relative clauses and requiring literal noun-phrase repetition for systemizing/autistic modes (PMC10123896), or enforcing mid-clause associative pivots for hyper-associative and flight-of-ideas modes (PMC10031728).